

Notice of Intent to Sole Source

Contracting Opportunities Announcement Type: Special Notice

Announcement Number: NIST-NOI-25-01053

Subject: Procurement of Integrated IMU-Based Human Motion Tracking System and EMG Measurement System

The National Institute of Standards and Technology (NIST) intends to negotiate, on a Sole Source basis, under the authority of FAR 6.302-1 with Noraxon USA Inc. located 15770 N GREENWAY HAYDEN LOOP, STE 100, SCOTTSDALE, AZ 85260 USA, for the procurement of the Integrated IMU-Based Human Motion Tracking System and EMG Measurement System. This equipment will be utilized by the Engineering Laboratory (EL), Sensing and Perception Systems Group at NIST Gaithersburg MD campus.

Sole source determination is based on the need to obtain a fully integrated system that meets all technical requirements, as specified below, to further the collection and analysis of data on human state while operating exoskeleton devices.

Line Item 0001: IMU-based human pose measurement system

Description: The system shall consist of a set of sensors sufficient for full-body motion tracking of one (1) user. The sensors must be able to be worn at the same time as an exoskeleton device. All accessories required for the charging of, communication with, and wearing of these sensors shall be provided with the system.

Quantity: One (1) system, consisting of multiple sensors

A. Technical Specifications

- a. The IMU-system must be able to measure rigid body positions with a static angular accuracy of less than 2 degrees.
- b. Each IMU device must have a battery life of at least 4 hours.
- c. The devices must be capable of recording data at a rate of at least 100hz.
- d. The system must be able to stream data from all sensors to an in-room computer for storage and processing. **(no cloud processing due to government restrictions)**

Line Item 0002: EMG measurement system

Description: The system shall consist of a set of sensors capable of measuring EMG signals at least eight (8) different locations on the body. The sensors must be able to be worn at the same time as an exoskeleton device. All accessories required for the charging of, communication with, and wearing of these sensors shall be provided with the system.

Quantity: One (1) system, consisting of multiple sensors

A. Technical Specifications

- a. The system must be sensitive enough to identify instances and relative magnitude of muscle effort (generally less than 2 μ V).
- b. Each IMU device must have a battery life of at least 4 hours.
- c. The devices must be capable of recording data at a rate of at least 100hz.
- d. The system must be able to stream data from all sensors to an in-room computer for storage and processing. **(no cloud processing due to government restrictions)**

Line Item 0003: Software License

Description: The delivered software must be capable of communicating with all sensors of both the EMG and IMU systems simultaneously. It must be able to, at minimum, record the raw data and perform preliminary processing to prepare it to be exported. It must be able to synchronize all collected data, and have the capacity to set up time-synchronization with third-party devices. All accessories required to successfully use the software with the sensing systems on a pre-existing computer must be included.

Quantity: One (1) license to the software

A. Technical Specifications

- a. The software must store and process **all data on the local computer**, not requiring sending of data to any cloud-based service.
- b. The software must be able to process imu data into a kinematic model of the human over time, tracking body pose and joint angles.

Based on market research, Noraxon USA Inc. has been identified as the only source capable of meeting all of NIST's minimum requirement.

*****The North American Classification System (NAICS) code for this acquisition is 334510 – Electromedical and Electrotherapeutic Apparatus Manufacturing with a small business size standard of 1250 Employees.

*****A determination by the Government not to compete the proposed acquisition based upon responses to this notice is solely within the discretion of the Government. Information received will normally be considered solely for the purpose of determining whether to conduct a competitive procurement.

*****Interested parties that believe they could satisfy the requirements listed above for NIST may clearly and unambiguously identify their capability to do so in writing on or before the response due date for this notice. This notice of intent is not a solicitation.

*****Any questions regarding this notice must be submitted in writing via email to Marvin Jean at marvin.jean@nist.gov . All responses to this notice of intent must be submitted so that they are received at marvin.jean@nist.gov no later than **May 15, 2025 at 12PM Eastern Standard Time.**

Each response should include the following Business Information:

- a. Contractor Name, Address
- b. Point of Contact Name, Phone Number, and Email address
- c. Contractor CAGE Code or UEI
- d. Contractor Business Classification (i.e, small business, 8(a), woman owned, HUBZone, veteran owned, etc.) as validated in System of Award Management (SAM). All offerors must register in SAM located at <https://sam.gov/content/home>
- e. Capability Statement